

Web Technology Theory Assignment

1. Introduction to HTML and its Structure

HTML (HyperText Markup Language) is the basic language used to create web pages. It defines the structure of a webpage using tags and elements. An HTML document mainly contains <html>, <head>, and <body> tags. The head section contains metadata and title, while the body contains visible content.

Key HTML Tags:

- <a> - Used to create hyperlinks.**
- <form> - Used to collect user input.**
- <table> - Displays data in rows and columns.**
- - Used to embed images.**
- , , - Used to create lists.**
- <p> - Defines a paragraph.**
-
 - Inserts a line break.**
- <label> - Defines labels for form inputs.**

2. Overview of CSS and Its Importance

CSS (Cascading Style Sheets) is used to style web pages. It controls layout, colors, fonts, and spacing. CSS makes websites attractive, responsive, and easy to maintain.

Types of CSS:

- Inline CSS - Applied directly inside HTML elements.**
- Internal CSS - Written inside a <style> tag in the head section.**
- External CSS - Written in a separate .css file and linked to HTML.**

3. Difference Between Margin and Padding

Margin creates space outside an element and separates it from other elements. Padding creates space inside an element between the content and the border.

4. CSS Pseudo-Classes

Pseudo-classes style elements based on their state.

- :hover - When mouse is over an element.**
- :focus - When input field is selected.**
- :active - When element is clicked.**

5. Difference Between ID and Class

ID is unique and used for one element (#).

Class is reusable and used for multiple elements (.).

6. Client-Server Architecture

In this model, the client sends requests and the server responds.

Client-side processing happens in the browser.

Server-side processing happens on the server.

Communication uses protocols like HTTP.

7. HTTP Protocol

HTTP (HyperText Transfer Protocol) is used for communication between client and server.

Client sends HTTP request and server sends HTTP response.

Headers contain information like request type and status codes.

8. J2EE and Multi-Tier Architecture

J2EE is used to build enterprise applications.

Web Tier – Handles client requests.

Application Tier – Contains business logic.

Database Tier – Manages data storage.

9. CGI (Common Gateway Interface)

CGI allows servers to execute programs to generate dynamic content.

Advantages: Simple and language independent.

Disadvantages: Slow and high server load.

10. Servlets

Servlets are Java programs that handle client requests.

Advantages: Fast, secure, platform independent.

Disadvantages: Complex and requires Java knowledge.

11. History of Servlets and Types

Servlets were introduced in 1997 by Sun Microsystems.

Types:

GenericServlet – Protocol independent.

HttpServlet – HTTP specific.

12. HttpServlet vs GenericServlet

GenericServlet supports all protocols.

HttpServlet supports HTTP methods like doGet and doPost.

13. Servlet Life Cycle

init() – Initialization

service() – Handles requests

destroy() – Cleanup

14. Creating and Configuring Servlets

Servlets are created by extending **HttpServlet**.

Configuration is done in **web.xml** with servlet name and URL mapping.

15. Logical URLs and ServletConfig

Logical URLs are user-friendly servlet mappings.

ServletConfig provides initialization parameters to servlets.

16. RequestDispatcher

Used to transfer request to another resource.

forward() – Transfers control.

include() – Includes response.

17. ServletContext and Listeners

ServletContext is application-wide.

Listeners monitor lifecycle events.

18. Filters

Filters intercept requests and responses.

Lifecycle: **init()**, **doFilter()**, **destroy()**

Configured in **web.xml**.

19. Session Management

Cookies, hidden fields, URL rewriting, and sessions are used.

Sessions are most secure for tracking users.